ALGEBRA 2 · LINEAR FUNCTIONS & SYSTEMS

Feasible Region Optimizer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Feasible Region Optimizer** on Mr. K's Classroom: Algebra 2 → Linear Functions & Systems → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Where in the shaded region do you think the best profit happens: the middle, an edge or a corner?

_____

_____

2 Explore use the tool and record what you see

Profit P = 30x + 20y (oven hours = 16)

Corner (x, y)	(0, 0)	(8, 0)	(6, 4)	(2, 8)	(0, 8)
Profit					

Change the profits

Profit function	Best corner	Best profit
P = 30x + 20y		
P = 10x + 20y		
P = 40x + 20y		

3 Write the rule fill in each blank from what you observed

1. The feasible region contains every point that satisfies _____ the constraints.
2. The maximum or minimum of the objective function happens at a _____ of the region.
3. When the profit line is parallel to an edge of the region, _____ corners can tie.
4. Sliding the profit line farther from the origin _____ the profit.

4 Check no tool: show your work

1. A region has corners (0, 0), (0, 6), (4, 4) and (7, 0). Maximize C = 5x + 3y.

2. Using the same corners, minimize C = 5x + 3y.

3. Is (3, 5) in the bakery region ($x + y \leq 10$, $2x + y \leq 16$, $y \leq 8$)?

4. Why don't we need to test points inside the region?

5 Reflect

Explain why the best value must be at a corner.